

ECE 4644 / 5610 Spring 2026 Mid-term Equation Sheet

Constants

$$\mu = 3.9860044 \times 10^5 \text{ km}^3/\text{s}^2 \quad \mu = GM_E \quad R_E = 6378.137 \text{ km} \quad G = 6.672 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

$$c = 3 \times 10^8 \text{ m/s} \quad T_S = 23 \text{ hr } 56 \text{ mn } 4.091 \text{ sc} \quad h_{\text{geo}} = 35,786 \text{ km}$$

$$T_o = 290 \text{ K} \quad k = 1.3810^{-23} \text{ W/K/Hz} = -228.6 \text{ dB W/K/Hz}$$

$$\text{Orbital elements} = (a, e, t_p, \Omega, \omega, i) \quad f [\text{GHz}] \times \lambda [\text{cm}] = 30$$

Equations

$$dA/dt = k \quad T_p = (2\pi a^{3/2})/\mu^{1/2} \quad r_{\text{apog}} = a(1 + e) \quad r_{\text{per}} = a(1 - e) \quad e = (a^2 - b^2)^{1/2}/a$$

$$r_o = \frac{a(1-e^2)}{1+e\cos(\phi_o)} \quad r_o = a(1 - e\cos E) \quad \eta = 2\pi/T \quad M = \eta(t - t_p) \quad M = E - e\sin(E)$$

$$x_o = r_o \cos(\phi_o) \quad y_o = r_o \sin(\phi_o)$$

$$\cos \gamma = \cos(L_e) \cos(l_s - l_e) \quad \text{El} = \tan^{-1} [(6.610745 - \cos \gamma) / \sin \gamma] - \gamma$$

$$\alpha = \tan^{-1}[(\tan|(l_s - l_e)|)/\sin |L_e|] \quad \text{SE: } Az = 180^\circ - \alpha \quad \text{SW: } Az = 180^\circ + \alpha \quad \text{NE: } Az = \alpha \quad \text{NW: } Az = 360^\circ - \alpha$$

$$f_{\text{doppler}} = v_T/\lambda$$

$$G = \frac{4\pi A_e}{\lambda^2} = \eta_A \frac{4\pi A_p}{\lambda^2} \quad G \simeq \frac{33,000}{\theta_H \theta_V} \quad G \simeq \frac{33,000}{\theta_B^2} \quad \theta_{3\text{dB}}^\circ \simeq 75\lambda/D$$

$$F = \frac{P_t G_t}{4\pi R^2} \quad P_r = F \times A_e \quad L_p = (4\pi R/\lambda)^2 \quad L_p = 10\log_{10}(4\pi R/\lambda)^2$$

$$P_r = P_t G_t G_r / L_p \quad P_r = P_t + G_t + G_r - L_p - L_{\text{tot}} \quad L_{\text{tot}} = L_1 + L_2 + \dots$$

$$N = kT_S B_n \quad N = -228.6 + T_S + B_n \quad T_S = T_{RF} + T_{in} + T_m/G_{RF} + T_{IF}/G_{RF} G_m$$

$$C/N = P_r/N = C - N = P_r - N \quad T_{no} = T_p(1 - G_l) \quad T_n = 290 \times (F - 1)$$

$$1/(C/N)_o = 1/(C/N)_{up} + 1/(C/N)_{dn}$$